

- 1 The diagram shows a prism $ABCDEFGH$ with a horizontal base.

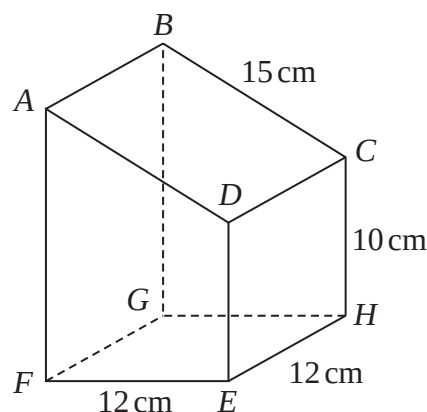


Diagram **NOT**
accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.

Trapezium $ADEF$ is a cross section of the prism where AF and DE are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between CF and the base $EFGH$.
Give your answer correct to one decimal place.

.....
(3)

- (b) Work out the length of BE .
Give your answer correct to one decimal place.

..... cm
(3)

(Total for Question 1 is 6 marks)

2 The diagram shows a cuboid $ABCDEFGH$.

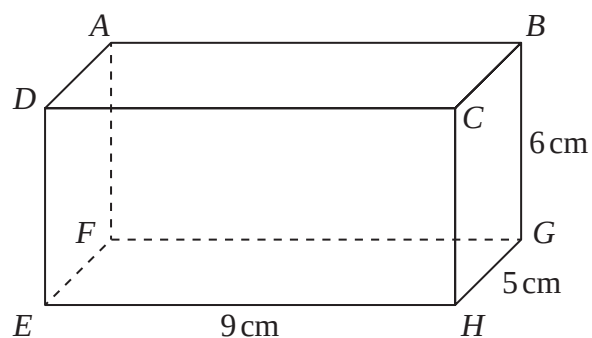


Diagram **NOT**
accurately drawn

$EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

(Total for Question 2 is 4 marks)

- 3 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

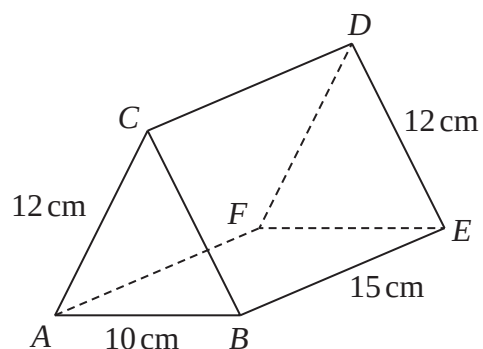


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

(Total for Question 3 is 4 marks)

- 4 ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B .

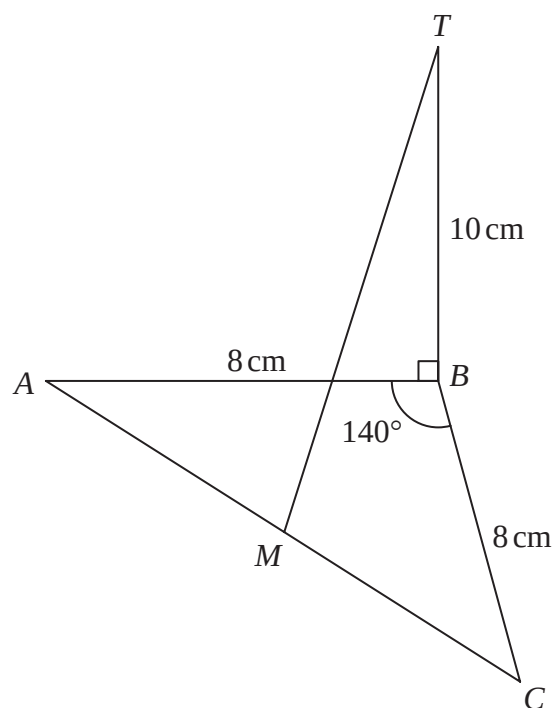


Diagram **NOT**
accurately drawn

$$\text{Angle } ABC = 140^\circ$$

$$AB = BC = 8 \text{ cm}$$

$$TB = 10 \text{ cm}$$

M is the midpoint of AC .

Calculate the size of the angle between MT and the horizontal plane ABC .
Give your answer correct to one decimal place.

o

(Total for Question 4 is 4 marks)

5 The diagram shows a solid prism $ABCDEFGH$.

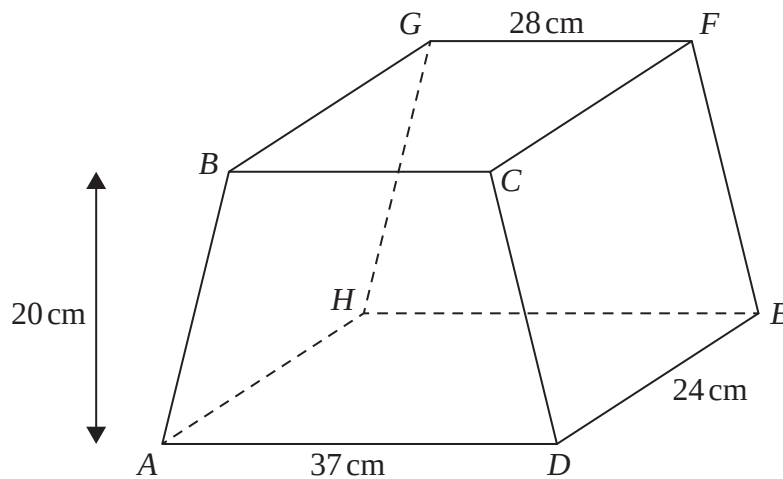


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37\text{ cm} \qquad GF = 28\text{ cm} \qquad DE = 24\text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm .

(b) Calculate the size of the angle between AF and the plane $ADEH$.

Give your answer correct to one decimal place.

(3)

(Total for Question 5 is 3 marks)

6 The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

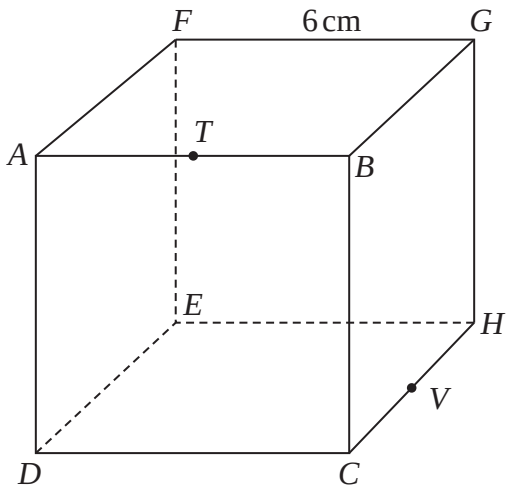


Diagram **NOT** accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

7 $ABCD$ is a horizontal rectangular field.

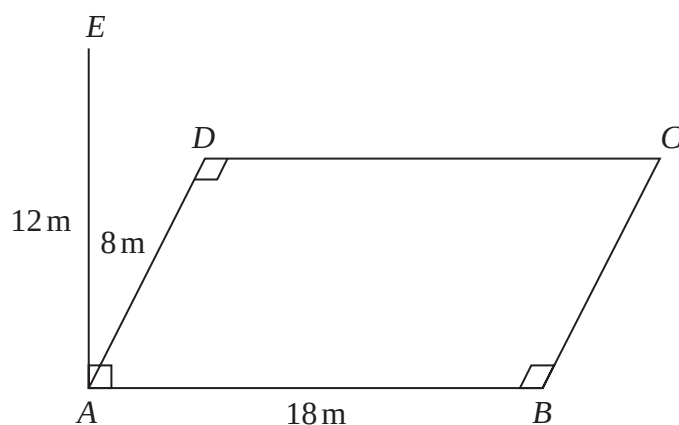


Diagram **NOT**
accurately drawn

A vertical pole, AE , is placed at the corner A of the field.

$$AE = 12 \text{ m} \quad AB = 18 \text{ m} \quad AD = 8 \text{ m}$$

Calculate the size of the angle between EC and the plane $ABCD$.
Give your answer correct to one decimal place.

(Total for Question 7 is 3 marks)

8 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

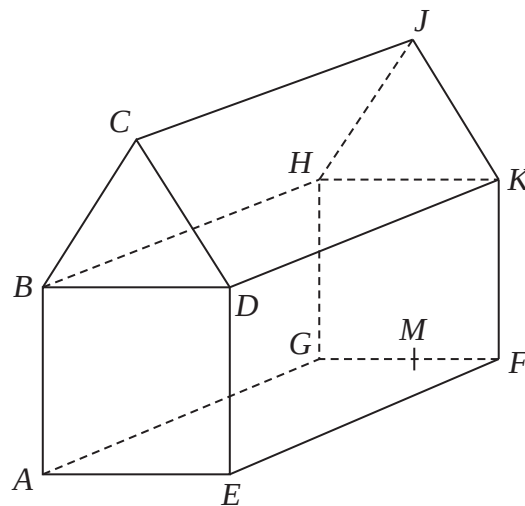


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where

$ABDE$ is a square

BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y^\circ$

Given that $\tan y^\circ = T$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$$T = \dots\dots\dots$$

(Total for Question 8 is 5 marks)

- 9 The diagram shows a triangular prism, $ABCDEF$, with a rectangular base $ABCD$

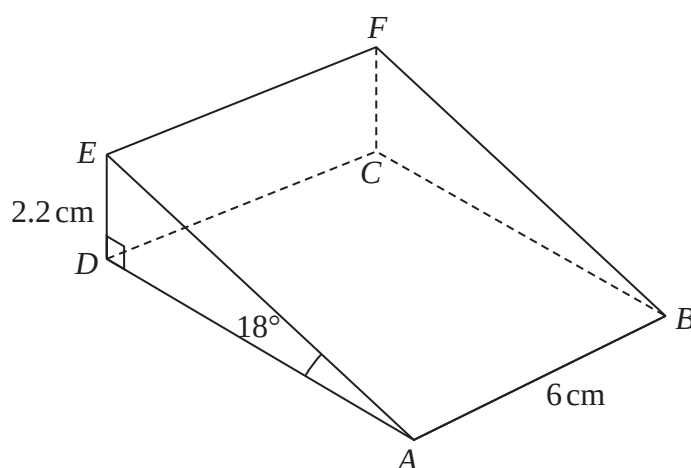


Diagram **NOT**
accurately drawn

$$AB = 6 \text{ cm}$$

$$DE = 2.2 \text{ cm}$$

$$\text{angle } DAE = 18^\circ$$

$$\text{angle } ADE = 90^\circ$$

Work out the angle that BE makes with the plane $ABCD$

Give your answer correct to one decimal place.

(Total for Question 9 is 4 marks)

10 The diagram shows a solid prism $ABCDEF GHIJ$

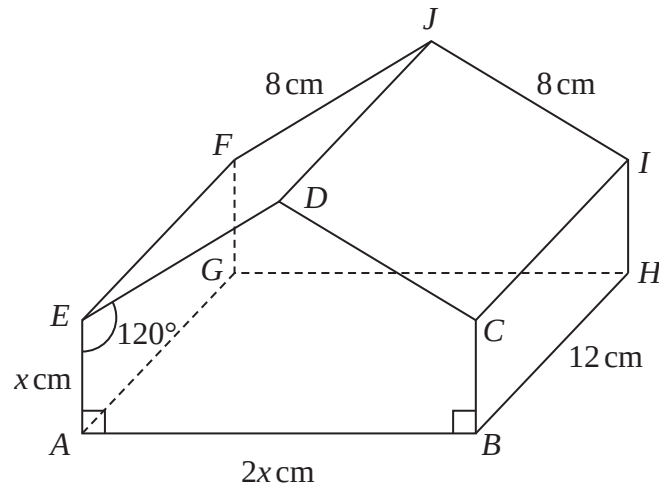


Diagram **NOT**
accurately drawn

The prism is such that each cross section is a pentagon where

$$AE = BC = x \text{ cm}$$

$$AB = 2x \text{ cm}$$

$$ED = CD = 8 \text{ cm}$$

$$\text{angle } EAB = \text{angle } CBA = 90^\circ$$

$$\text{angle } AED = \text{angle } BCD = 120^\circ$$

Given that $AG = BH = EF = DJ = CI = 12 \text{ cm}$

calculate the angle that AJ makes with the base $ABHG$ of the prism.

Give your answer correct to 3 significant figures.

o

(Total for Question 10 is 5 marks)

11 Here is a cuboid $PQRSTUVW$

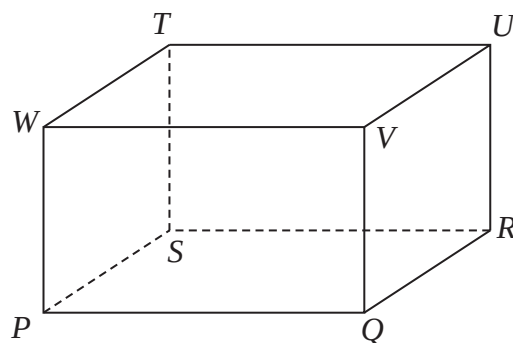


Diagram **NOT**
accurately drawn

$$PR = 42 \text{ cm}$$

The size of the angle between PU and the plane $PQRS$ is 30°

M is the midpoint of PR

- (b) Work out the size of angle UMR
Give your answer correct to 3 significant figures.

(3)

(Total for Question 11 is 3 marks)

12 The diagram shows a triangle ABC and a flagpole BF

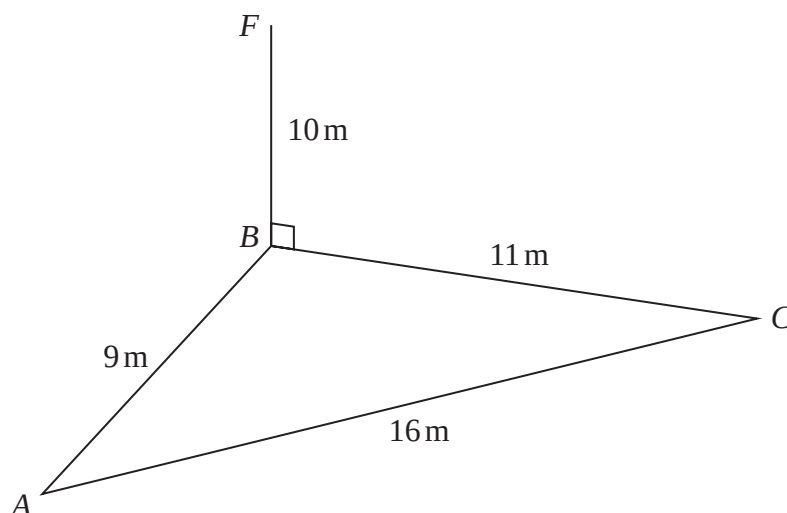


Diagram **NOT**
accurately drawn

A , B and C are points on horizontal ground.

BF is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

D is the point on AC such that angle $BDC = 90^\circ$

Work out the size of the angle of elevation of the point F from the point D
Give your answer correct to one decimal place.

o

(Total for Question 12 is 5 marks)
